

Practice of Epidemiology

Iterative Causal Forest: A Novel Algorithm for Subgroup Identification

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Precisely and efficiently identifying subgroups with heterogeneous treatment effects (HTEs) in real-world evidence studies remains a challenge. Based on the causal forest (CF) method, we developed an iterative CF (iCF) algorithm to identify HTEs in subgroups defined by important variables. Our method iteratively grows different depths of the CF with important effect modifiers, performs plurality votes to obtain decision trees (subgroup decisions) for a family of CFs with different depths, and then finds the cross-validated subgroup decision that best predicts the treatment effect as a final subgroup decision. We simulated 12 different scenarios and showed that the iCF outperformed other machine learning methods for interaction/subgroup identification in the majority of scenarios assessed. Using a 20% random sample of fee-for-service Medicare beneficiaries initiating sodium-glucose cotransporter-2 inhibitors or glucagon-like peptide-1 receptor agonists, we implemented the iCF to identify subgroups with HTEs for hospitalized heart failure. Consistent with previous studies suggesting patients with heart failure benefit more from sodium-glucose cotransporter-2 inhibitors, iCF successfully identified such a subpopulation with HTEs and additive interactions. The iCF is a promising method for identifying subgroups with HTEs in real-world data where the potential for unmeasured confounding can be limited by study design.

causal forest; heterogeneous treatment effect; iterative causal forest; pharmacoepidemiology; precision medicine; subgroup identification

Abbreviations: CATE, conditional average treatment effect; CF, causal forest; CHF, congestive heart failure; CVD, cardiovascular disease; GLP1RA, glucagon-like peptide-1 receptor agonists; HFF, hospitalized heart failure; HTE, heterogeneous treatment effect; iCF, iterative causal forest; IT, interaction tree; LASSO, least absolute shrinkage and selection operator; ML, machine learning; oneCF, one causal forest; PS, propensity score; RD, risk difference; ROWSi, regularized outcome weighted subgroup identification; SGLT2i, sodium-glucose cotransporter-2 inhibitors.

Average treatment effects are likely to vary across subpopulations (1–8). Real-world evidence studies are often larger and more inclusive than randomized trials, which provides an opportunity to assess heterogeneous treatment effects (HTEs) (9, 10). Traditional subgroup analyses looking at one factor at a time may mask true HTEs, thus it is important to identify subpopulations defined by multiple covariates (4–8). Subgroup identification commonly falls into 2 frameworks: identifying subpopulations that may benefit or be harmed more by a given treatment, and determining the rules for treatment assignment. In this study, we focus on the first framework (11, 12).

Traditional methods identify subgroups by fitting a parametric outcome model, which includes both main effects and prespecified treatment-covariate interactions and assesses heterogeneity based on the statistical significance of such interactions; however, this model is limited by prior knowledge, *P* value use for decision-making, and model misspecification (13). Tree-based machine learning (ML) subgrouping methods based on nonparametric approaches do not have such limitations (13). A tree can be expressed by designating one node as the root node and directing all edges away from it; terminal nodes at the bottom of a tree are leaves (i.e., subgroups), and the depth of a tree is the

length of its longest path (14) (Web Figure 1, available at <https://doi.org/10.1093/aje/kwad219>). A decision tree (using a tree structure to summarize splitting rules), which recursively partitions the sample into subgroups defined by covariates (splitting variables) to minimize heterogeneity within subgroups, is a tool to identify subgroups and treatment-covariate interactions (14–16). For example, the interaction tree (IT) follows the classification and regression tree algorithm and builds splitting rules based on treatment-covariate interaction tests (17).

A random forest is a large collection of decorrelated trees (each tree draws a subsample from the overall sample and randomly selects part of features to split) and works by averaging predictions of individual trees (majority vote for categorical outcomes), which is a popular ML method for fitting flexible prediction models (18). For instance, the aVirtualTwins method works by first predicting treatment differences based on random forests then applying trees to those estimates to form subgroups (19). However, these methods are not designed for observational data or estimating causal effects (20). Recently, Wager and Athey extended the random forest algorithm and developed the causal forest (CF) for estimating HTEs in a causal framework in observational data (21–25). As the CF was originally developed for HTE estimation rather than subgroup identification, a decision tree from a CF often splits the study population by many features, leading to trivial subgroups, and is unstable due to its hierarchical nature (i.e., the effect of an error in the top split is spread to all splits below it) (16). Thus, efficiently identifying important subgroups using real-world data to inform clinical decisions remains a challenge.

In this study, we used decision trees from a CF and implemented a few extra steps to: 1) iteratively grow different depths of the CF; 2) select a decision tree that best represents the HTE signal from each forest, then votes to select the most stable tree structure as a subgroup decision (the set of leaves in a tree) at each depth; 3) build models to predict treatment effect using subgroup decisions; and 4) find the cross-validated subgroup decision that most accurately predicts the treatment effect to discover important and stable subgroups. We call this ML subgrouping algorithm the “iterative causal forest” (iCF).

METHODS

Causal forest

A CF assesses HTEs in a causal framework under the “no unmeasured confounding” assumption. CFs can be used to estimate conditional average treatment effects (CATEs, i.e., per-subgroup average treatment effects), which can be conceptualized as an estimator of parameters of a penalized model (whose loss function is referred as “R-loss”) (24–27). The CF splits a population to maximize the heterogeneity of the CATEs across subgroups, and the “best tree” procedure aims to select a single tree from a CF that represents the heterogeneous signal in the best way possible (i.e., with the smallest root-node impurity R-loss) (28, 29) (Web Appendixes 1 and 2). We refer readers to the original papers (21, 22) for details regarding the CF and best tree.

Proposed method: iterative causal forest

Following the approach suggested by Athey et al. (22, 24), in the preliminary stage (Web Appendix 3, Figure 1), we first trained 2 regression forests individually using all covariates $\mathbf{X}$ to predict the expected outcome ($\mathbb{E}[Y_i | \mathbf{X}_i = \mathbf{x}]$) and propensity score ($PS = \mathbb{P}[W_i = 1 | \mathbf{X}_i = \mathbf{x}]$; W denotes treatment) (other predictive methods such as logistic regression could also be used), which are plugged into a centered (residualized) form of a partially linear model to estimate causal effect and eliminate measured confounding (Web Appendix 1) (24). Then, we used the predictions as input to grow a raw CF (a “pilot” CF using all covariates without tuning parameters) to perform a homogeneity test and calculate the “variable importance” value (a weighted sum of how many times a variable was split at each depth in a CF) (24). Next, if the homogeneity test was significant at P value $\leq \alpha$ (e.g., at 0.1), we continued the following steps in an f -fold cross-validation (e.g., $f = 10$).

A family of subgroup decisions from a pruned iCF on a training set

Growing a shallow CF by key treatment effect modifiers allows us to target important and interpretable subgroups (18), rather than targeting possible heterogeneity. We use selected features $\mathbf{X}_S$ with the “variable importance” value greater than the mean (i.e., greater than the mean measure of “importance” for features (24)) to tune and find the minimum leaf size (i.e., the minimum sample size for each subgroup) leading to a different depth (D) of CF, $D \subseteq \{2, 3, 4, 5\}$ (Web Appendix 3, Figure 1).

Next, we iteratively grew a total of K CFs to get K best trees (each best tree denoted by $\hat{L}_{BT,D,k}$) at different depths, individually, resulting in a collection of M different best trees ($M \leq K$). We then picked the most popular best tree structure by plurality vote (16), ignoring splitting values, let $\hat{L}_D = \text{plurality vote } \{\hat{L}_{BT,D,k}\}_1^K$. As a best tree is obtained by selecting the tree with the smallest root-node impurity R-loss, among K structures of best trees $\hat{L}_{BT,D,k}$, we intuited that a structure that appears most often is more likely to represent true heterogeneity. We then assigned the mean splitting value from the most popular tree structures to each splitting node of $\hat{L}_D$ to obtain a synthetic decision tree (e.g., we split samples by $X_2 \leq 1.5$ and $X_2 > 1.5$, where 1.5 is the mean splitting value for X_2 from the voted best trees), leading to the subgroup decision G_D (i.e., the set of groups (G) selected for a given depth; each group g_D is defined by indicator variables $\mathbf{X}_i$). A depth $= d$ tree could identify a d -way interaction (i.e., $(d-1)$ X_i interacting with W) (Web Figure 1). For example, for a 2-way interaction of W and binary X_3 denoted by $W:X_3$, the best tree of a $D2$ CF is likely to split by X_3 and lead to G_2 : 1) $X_3 = 0$; 2) $X_3 = 1$, suggesting X_3 interacting with W .

A family of transformed outcome models based on G_D on a validation set

As the goal is to pick the optimal grouping among different G_D on the validation set, we used a transformed

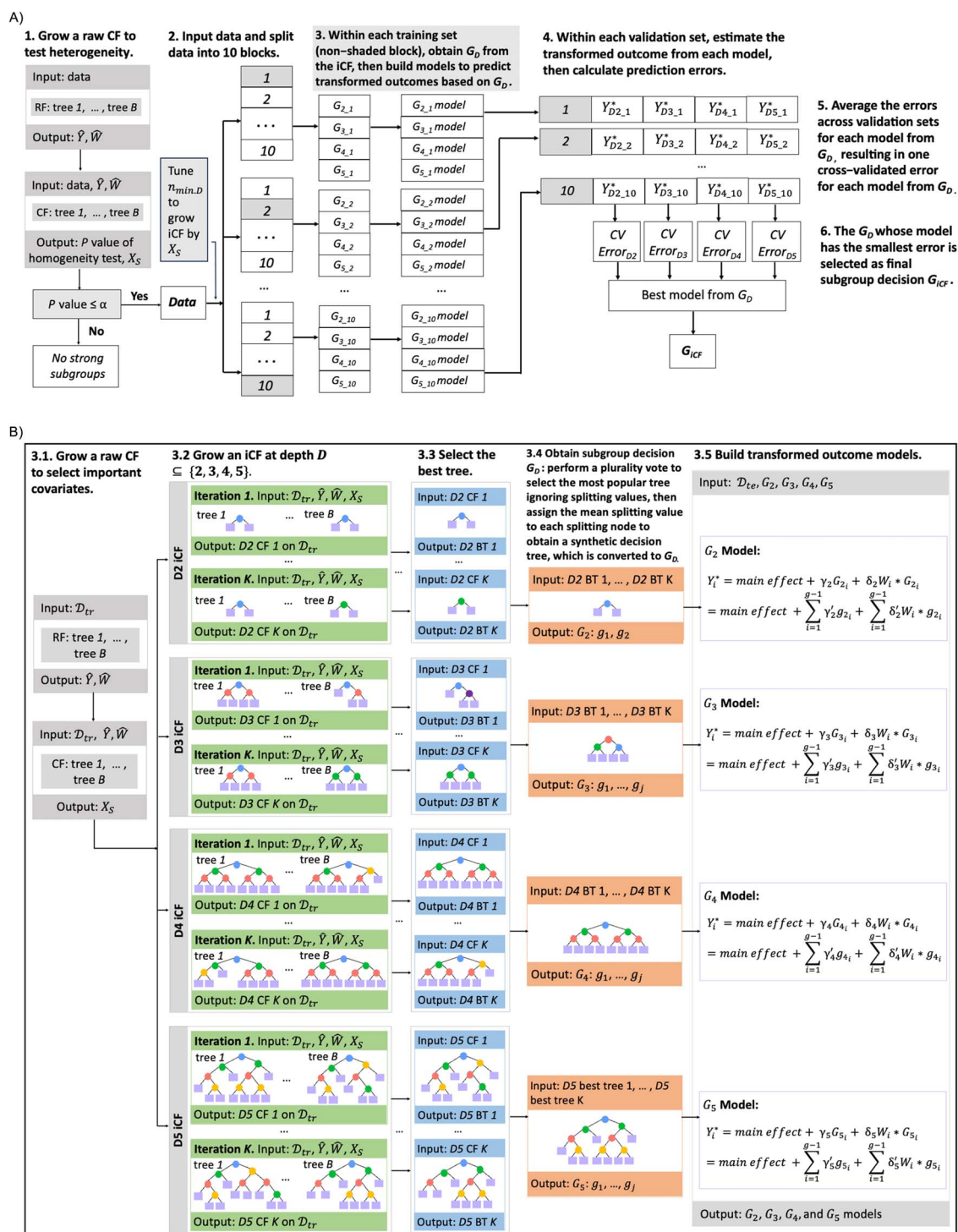


Figure 1. Iterative causal forest (iCF) algorithm. A) iCF workflow. If the homogeneity test from a raw casual forest (CF) is significant at P value $\leq \alpha$ (e.g., at 0.1), then divide the data into 10 groups of equal size. The first group is treated as the testing set, and the remaining 9 groups as the training set. Repeat the procedure 10 times. B) Step 3 of iCF workflow (obtain a family of subgroup decisions from a pruned iCF on the training set and a family of transformed outcome models based on subgroup decisions (G_D) on the test set). With a cross-validation approach, each G_D selected will be noted as $G_{D,f}$, where f denotes the fold of cross-validation (e.g., $f = 10$). If $G_{D,f}$ varies across training sets, then perform plurality vote across f subgroup decisions to select the most popular one and use its corresponding model. Then, average the error across validation sets for each model from G_2, G_3, G_4 , and G_5 , individually, and the G_D with a corresponding model with the smallest cross-validated error is selected as the final subgroup decision G_{ICF} . BT, best tree; CV, cross-validation; $\mathcal{D}_{te}$, test data; $\mathcal{D}_{tr}$, training data; $n_{\min,D}$, minimum leaf size; g_j , individual group of G_D ; PS, propensity score; RF, random forest; $\hat{W}$, predicted propensity score; X_S , selected features; $\hat{Y}$, predicted outcome; Y^* , predicted transformed outcome.

Table 1. The 12 Simulated Scenarios for Heterogeneous Treatment Effect^a

Scenario	Order of Interaction	Treatment: Covariates Interaction	Type of Variables Related to HTE	β Parameters ^b
1	N/A	N/A	N/A	All β parameters equal to 0 (no HTE)
2	2-way	W:X ₃	X ₃ binary	Scenario 1 + β ₃ ≠ 0
3	2-way	W:X ₂	X ₂ continuous	Scenario 1 + β ₂ ≠ 0;
4	2-way	W:X ₂	X ₂ ordinal ^c	Scenario 1 + β ₂ ≠ 0
5	Two 2-way	W:X ₃ & W:X ₁	X ₃ binary; X ₁ binary	Scenario 1 + β ₁ ≠ 0 & β ₃ ≠ 0
6	3-way	W:X ₃ :X ₁	X ₃ binary; X ₁ binary	Scenario 1 + β _{1,3} ≠ 0 & β ₁ ≠ 0 & β ₃ ≠ 0
7	3-way	W:X ₃ :X ₂	X ₃ binary; X ₂ continuous	Scenario 1 + β _{2,3} ≠ 0 & β ₂ ≠ 0 & β ₃ ≠ 0
8	3-way	W:X ₃ :X ₂	X ₃ binary; X ₂ ordinal ^c	Scenario 1 + β _{2,3} ≠ 0 & β ₂ ≠ 0 & β ₃ ≠ 0
9	4-way	W:X ₃ :X ₁ :X ₈	X ₃ binary; X ₁ binary; X ₈ binary;	Scenario 1 + β _{1,3,8} ≠ 0 & β ₁ ≠ 0 & β ₃ ≠ 0 & β ₈ ≠ 0
10	4-way	W:X ₃ :X ₁ :X ₂	X ₃ binary; X ₁ binary; X ₂ continuous	Scenario 1 + β _{1,2,3} ≠ 0 & β ₁ ≠ 0 & β ₂ ≠ 0 & β ₃ ≠ 0
11	4-way	W:X ₃ :X ₁ :X ₂	X ₃ binary; X ₁ binary; X ₂ ordinal ^c	Scenario 1 + β _{1,2,3} ≠ 0 & β ₁ ≠ 0 & β ₂ ≠ 0 & β ₃ ≠ 0
12	5-way	W:X ₃ :X ₁ :X ₈ :X ₉	X ₃ binary; X ₁ binary; X ₈ binary; X ₉ binary	Scenario 1 + β _{1,3,8,9} ≠ 0 & β ₁ ≠ 0 & β ₃ ≠ 0 & β ₈ ≠ 0 & β ₉ ≠ 0

Abbreviations: HTE, heterogeneous treatment effect; N/A, not applicable.

^a Across all simulations, the outcome was generated according to the following model (where individual indexes are suppressed for clarity, so that $Y = Y_i$, for example):

$$Y = -3.85 + 0.3X_1 - 0.36X_2 - 0.73X_3 - 0.2X_4 + 0.71X_8 - 0.19X_9 + 0.26X_{10} - 0.4W + \beta_1 W^* X_1 + \beta_2 W^* I(X_2 > \text{cutoff}) + \beta_3 W^* X_3 + \beta_8 W^* X_8 + \beta_9 W^* X_9 + \alpha_{1,2} X_1^* X_2 + \alpha_{1,3} X_1^* X_3 + \alpha_{1,8} X_1^* X_8 + \alpha_{2,3} I(X_2 > \text{cutoff})^* X_3 + \alpha_{3,8} X_3^* X_8 + \alpha_{1,9} X_1^* X_9 + \alpha_{8,9} X_8^* X_9 + \alpha_{3,9} X_3^* X_9 + \beta_{1,2} W^* X_1^* X_2 + \beta_{1,3} W^* X_1^* X_3 + \beta_{1,8} W^* X_1^* X_8 + \beta_{2,3} W^* I(X_2 > \text{cutoff})^* X_3 + \beta_{3,8} W^* X_3^* X_8 + \beta_{1,2,3} W^* X_1^* I(X_2 > \text{cutoff})^* X_3 + \beta_{1,3,8} W^* X_1^* X_3^* X_8 + \beta_{1,3,9} W^* X_1^* X_3^* X_9 + \beta_{3,8,9} W^* X_3^* X_8^* X_9 + \beta_{1,8,9} W^* X_1^* X_8^* X_9 + \beta_{1,3,8,9} W^* X_1^* X_3^* X_8^* X_9 + \epsilon$$

where ϵ is a normal distribution with mean 0 and standard deviation 1.

^b The values of β and α parameters are shown in Web Table 1.

^c For scenarios 4, 8, and 11, X₂ is an ordinal variable with 4 levels (0, 1, 2, 3).

version of the outcome constructed from inverse-probability of treatment weight (IPTW): $Y_i^* = Y_i/PS_i$ if $W_i = 1$, $Y_i^* = -Y_i/(1 - PS_i)$ if $W_i = 0$ ($CATE = \mathbb{E}[Y_i(\frac{W_i}{PS_i} - \frac{1-W_i}{1-PS_i}) | \mathbf{X}_i = \mathbf{x}]$) to focus on treatment effect rather than outcomes (23). Here, PS_i can be reestimated by a regression forest or alternative methods such as logistic regression with least absolute shrinkage and selection operator (LASSO) penalty (30). Then, we fitted 4 prediction models to the transformed outcomes for different G_D , respectively, including W , $\mathbf{X}_i$, G_D as a factor variable with q levels (for q subgroups, e.g., G_2 , has 2 levels, 0 and 1, for 2 groups), and the $W:G_D$ interaction (as a proxy of all interactions between W and $\mathbf{X}_i$) (Web

Appendix 3):

$$Y_i^* = \beta_0 + \sum_j \beta_{ij} X_{ij} + \alpha W_i + \gamma_D G_{Di} + \delta_D W_i^* G_{Di} \\ = \beta_0 + \sum_j \beta_{ij} X_{ij} + \alpha W_i + \sum_{i=1}^{g-1} \gamma_D' g_{Di} \\ + \sum_{i=1}^{g-1} \delta_D' W_i^* g_{Di}$$

where γ_D and δ_D are coefficients for G_D and $W:G_D$, respectively, and γ_D' and δ_D' are coefficients for individual group g

of G_D and the $W:g_D$ interaction, respectively; $(g - 1)$ stands for the total number of interaction terms when one group (level) will be dropped, so the model will not be overparameterized. Note, the g and $W:g_D$ terms drive the differences in model performance (the main effects are the same for the four models), and our goal is to pick the best model rather than to predict Y_i^* . Thus, model misspecification is unlikely to affect the robustness of our algorithm. The model with the smallest root mean-squared error is selected as the best model among the 4. The G_D in the selected best model will be chosen as the subgroup decision G_{iCF} from the iCF.

We proposed to use the above procedures in an f -fold cross-validation to obtain a family of models to predict treatment effect in validation sets (the f fold) based on the G_D obtained from shallow CFs developed in training sets (the $f - 1$ folds), then use the G_D from the best model (with the smallest cross-validated root mean-squared error among the three) as the final subgroup decision G_{iCF} .

iCF tuning parameters

The iCF includes tuning parameters from its three component algorithms, the random forest, CF, and best tree. In the preliminary stage, we used the default parameters from the R function `regression_forest` (for predicting outcome and PS) and `causal_forest` (for developing a raw CF) from `grf` package v2.2.0 (22). To grow the iCF, we used the `causal_forest` and `find_best_tree` default parameters from the R function (29) and $\alpha = 0.1$ for the homogeneity test. We tuned only for the number of trees ($ntree$), the iteration number (K , as a reasonable K leads to stable results and is not computationally expensive), and minimum leaf size ($n_{min.D}$).

Simulation studies

To test the performance of the iCF under various scenarios with no unmeasured confounding, we used a similar simulation setup as initially described by Setoguchi et al. (31) and implemented/refined by others (32–33). The sample size was $n = 20,000$ (simulations were repeated for $n = 10,000$ and $5,000$, separately). We simulated 4 standard normal (X_2, X_4, X_7, X_{10}) and 6 binary ($X_1, X_3, X_5, X_6, X_8, X_9$) covariates and assigned a dichotomous treatment W via the simple (main effect only) logistic PS model modeling the propensity of receiving a statin in actual claims data (31):

$$p(W | X_1, \dots, X_7) = (1 + \exp(-(0.8X_1 - 0.25X_2 + 0.6X_3 - 0.4X_4 - 0.8X_5 - 0.5X_6 + 0.7X_7)))^{-1} \text{ resulting in a marginal } p(W) \sim 0.5.$$

Among the 10 covariates, 4 are confounders (X_1 – X_4), 3 predict the outcome only (X_8 – X_{10}), and 3 predict the treatment only (X_5 – X_7); a causal diagram is given in Web Figure 2. We set a cutoff value of 0.5 for continuous variable X_2 to define subgroups (the correct tree split on X_2 at 0.5) for the $W:X_2$ interaction, which makes it possible to identify true simulated values of HTEs by the continuous covariates using recursive partitioning. In a second set of simulations, we converted X_2 to an ordinal variable with 4 levels (0, 1, 2, and 3) with a cutoff value of 1 to define subgroups (the correct tree splits on X_2 at 1). We simulated the outcome

for each individual from a normal distribution with a mean of $E[Y | X]$ and a standard deviation of 1 (33). Across all simulations, the outcome was generated according to the following model (where individual indexes are suppressed for clarity, so that $Y = Y_i$, for example):

$$\begin{aligned} Y = & -3.85 + 0.3X_1 - 0.36X_2 - 0.73X_3 - 0.2X_4 + 0.71X_8 \\ & - 0.19X_9 + 0.26X_{10} - 0.4W + \beta_1 W^*X_1 + \beta_2 W^*I(X_2 > \text{cutoff}) \\ & + \beta_3 W^*X_3 + \beta_8 W^*X_8 + \beta_9 W^*X_9 + \alpha_{1,2}X_1^*X_2 \\ & + \alpha_{1,3}X_1^*X_3 + \alpha_{1,8}X_1^*X_8 + \alpha_{2,3}I(X_2 > \text{cutoff})^*X_3 \\ & + \alpha_{3,8}X_3^*X_8 + \alpha_{1,9}X_1^*X_9 + \alpha_{8,9}X_8^*X_9 + \alpha_{3,9}X_3^*X_9 \\ & + \beta_{1,2}W^*X_1^*X_2 + \beta_{1,3}W^*X_1^*X_3 + \beta_{1,8}W^*X_1^*X_8 \\ & + \beta_{2,3}W^*I(X_2 > \text{cutoff})^*X_3 + \beta_{3,8}W^*X_3^*X_8 + \\ & \beta_{1,2,3}W^*X_1^*I(X_2 > \text{cutoff})^*X_3 + \beta_{1,3,8}W^*X_1^*X_3^*X_8 \\ & + \beta_{1,3,9}W^*X_1^*X_3^*X_9 + \beta_{3,8,9}W^*X_3^*X_8^*X_9 + \beta_{1,8,9}W^*X_1^* \\ & X_8^*X_9 + \beta_{1,3,8,9}W^*X_1^*X_3^*X_8^*X_9 + \epsilon \end{aligned}$$

where HTEs by covariates are given by the β parameters (values shown in Web Table 1). The error ϵ is a normal distribution with a mean of 0 and a standard deviation of 1. From this model, we simulated 12 scenarios: no HTEs (scenario 1) and, 2-way, two 2-way, 3-way, 4-way, and 5-way interactions (scenarios 2–4, 5, 6–8, 9–11, and 12, respectively). We reexamined scenarios 3, 7, and 10 in scenarios 4, 8, and 11, respectively, by replacing continuous X_2 with ordinal X_2 (Table 1).

For each scenario, we simulated 500 data sets for analysis. In each data set, data were split such that 80% were used as a training set to learn the interactions/subgroups, and 20% were used as a testing set for estimating the CATEs for the subgroups learned in the training set. We compared the iCF's performance to that of other subgrouping methods: 1) IT (17); 2) linear regression with LASSO penalty to identify interactions; 3) FindIt, an ML method that identifies heterogeneity in causal effects through variable selection by developing a squared loss support vector machine with 2 separate LASSO constraints (20); 4) regularized outcome weighted subgroup identification (ROWSi), which approximates a target function reflecting treatment-covariate interactions whose values directly reflect correct treatment assignment for patients (13); and 5) one CF without iterations (oneCF). For the iCF, we set $ntree = 200$ and $K = 50$; implementation details for the other methods are shown in Web Appendix 4.

Interaction identification. For the tree-based methods, interactions were identified by converting subgroup decisions into interactions and ignoring splitting values, for example, the subgroup decision G_2 : 1) $X_2 \leq 0.3$; 2) $X_2 > 0.3$ leads to $W:X_2$ interaction. For the other methods (LASSO, FindIt, and ROWSi), realizing that weak interactions may be identified due to chance (21, 34), we extracted interactions from the models by requiring the absolute values of their coefficients be greater than the mean absolute value of coefficients for all interactions, so we could focus on important treatment modifiers. We considered the interaction identification to be accurate if the highest order of interaction was correctly identified and others were fully/partially identified (e.g., in scenario 5, if a method identified $W:X_1:X_3$ but missed $W:X_3$, we considered it

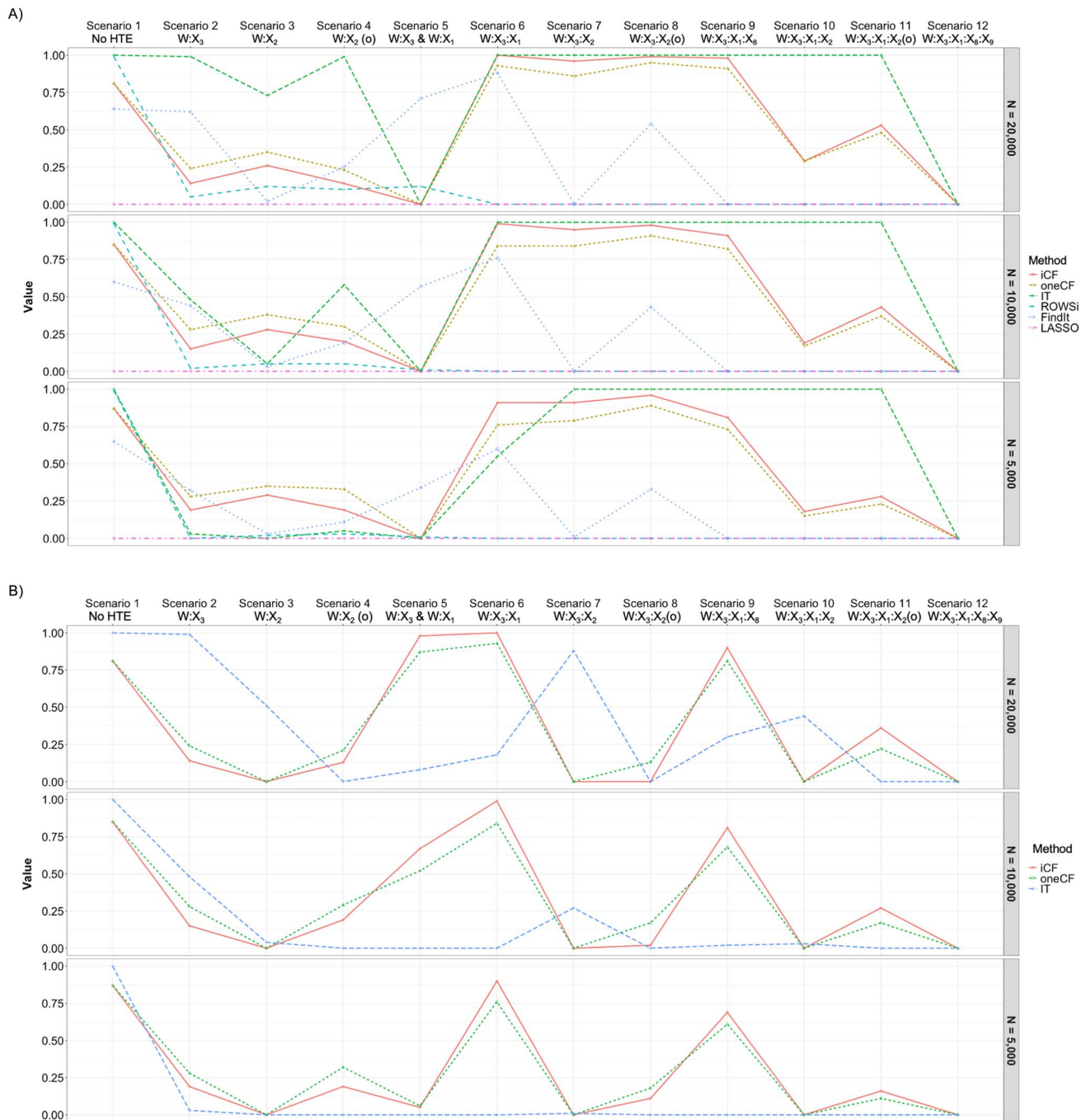


Figure 2 Continues

accurate). We evaluated the performance of each method for its sensitivity (defined as the proportion of true interactions being accurately identified) and specificity (defined as the proportion of a true zero interaction being identified as a zero interaction) across simulations.

Subgroup identification. We assessed the performance of subgroup identification only for the tree-based methods, which explicitly provide subgroup decisions. We defined a

“true,” nonoverlapping subgroup decision from a plausible tree structure based on true interaction in each scenario (e.g., in scenario 6, a tree may split on X_3 first then on X_1 , resulting in the true subgroup decision G_3 with 4 groups: 1) $X_3 = 0$, $X_1 = 0$; 2) $X_3 = 0$, $X_1 = 1$; 3) $X_3 = 1$, $X_1 = 0$; and 4) $X_3 = 1$, $X_1 = 1$). True subgroups across different scenarios are given in Web Table 1. We considered the subgroup identification accurate if it met 3 conditions: 1) both splitting variables and splitting values (rounded to 1 decimal place for continuous

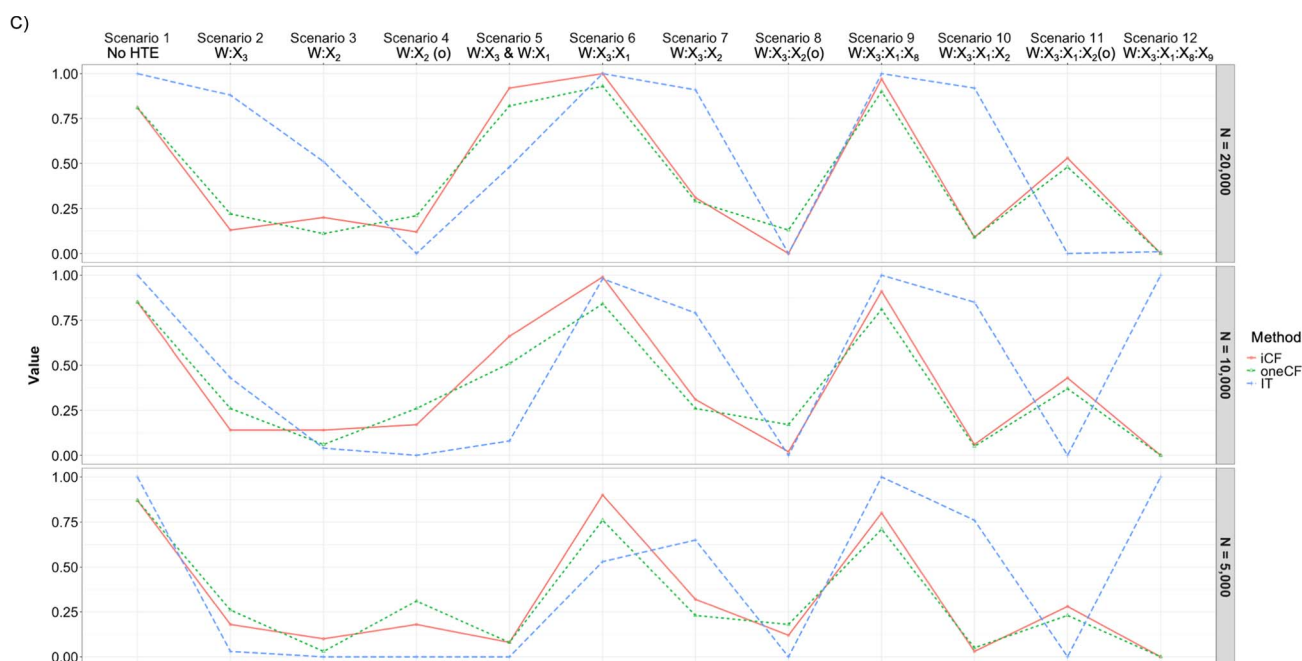


Figure 2. Performance of the iterative causal forest (iCF) and other machine learning methods. A) The performance for interaction identification; B) the performance for subgroup identification; C) the performance for identifying the subgroup with the largest conditional average treatment effect. The value is for specificity in scenario 1 and for sensitivity in scenarios 2–12. $X_2(o)$ is an ordinal variable with 4 levels (0, 1, 2, 3). For scenarios involving interacting with continuous variable X_2 , we set a cutoff value of 0.5 for X_2 to define subgroups and the correct tree split on X_2 at 0.5. For scenarios involving interacting ordinal variable X_2 , we define a cutoff value of 1 for ordinal variable X_2 , so that heterogeneous treatment effects (HTEs) differs depending on whether $X_2 \leq 1$ or not, and the correct tree split on X_2 is at 1. IT, interaction tree; LASSO, least absolute shrinkage and selection operator; oneCF, one causal forest; ROWSi, regularized outcome weighted subgroup identification.

variables with a standard deviation of 1) are accurate for each subgroup; 2) all subgroups are accurately identified; and 3) the number of subgroups is accurate. We assessed performance based on sensitivity (the proportion of true subgroups being accurately identified) and specificity (the proportion of true zero interactions being estimated as no subgroups) across simulations.

Identifying the subgroup with the largest CATE. In the testing set, we evaluated whether a method could identify the subgroup with the largest enhanced treatment effect (19), which is defined by the subgroup with the largest absolute difference between the CATE and average treatment effect, that is:

$$Q(A) = |E[Y|W = 1, \mathbf{X} \in g] - E[Y|W = 0, \mathbf{X} \in g]| - (E[Y|W = 1] - E[Y|W = 0])|.$$

To calculate the CATE, the PS is computed by a logistic regression model with the LASSO penalty (Web Appendix 4). We then used the stabilized IPTW (using the weights of $1/PS$ multiplied by the marginal proportion of W for the treatment group and $1/(1 - PS)$ multiplied by the marginal proportion of $(1 - W)$ for the comparator group (9)) to adjust for confounding. We assessed performance based on sensitivity (the proportion of the subgroup with the largest $Q(A)$ being estimated as the largest $Q(A)$) and specificity (the

proportion of true zero interactions being estimated as no HTEs) across simulations.

Empirical example

We compared the 2-year risk difference (RD) of hospitalized heart failure (HHF) of initiating any sodium-glucose cotransporter-2 inhibitors (SGLT2i) versus glucagon-like peptide-1 receptor agonists (GLP1RA) using a 20% random sample of all fee-for-service US Medicare beneficiaries who had parts A (inpatient), B (outpatient physician services), and D (dispensed prescription drugs) coverage for at least 1 month from January 2012 to December 2017. Large-scale, placebo-controlled randomized trials have shown that SGLT2i reduce HHF risk, while GLP1RA's effects on HHF are still not clear (35–44). The active-comparator, new-user study design, inclusion/exclusion criteria, diagnosis and procedure codes for outcome and covariate definition, and statistical analysis have been described previously (45). We quantified the treatment effect by the RD (46) for HHF between the SGLT2i and GLP1RA groups ($\Delta Y = \bar{Y}_{treated} - \bar{Y}_{untreated}$) in an initial treatment analysis over a maximum 2-year follow-up period (45) requiring patients to enter the cohort earlier than January 1, 2016 (to increase the probability for a 2-year follow-up).

We used the same tuning parameters of the iCF for the simulation except $n_{tree} = 2,000$ and $K = 1,000$ for

Table 2. Crude and Adjusted Risk Differences for Hospitalized Heart Failure Associated With the Use of SGLT2i vs. GLP1RA According to Initial Treatment Analysis Over a Maximum 2-Year Follow-up Period by Subgroup Decisions From the Iterative Causal Forest^a, Using Data From Medicare Claims, United States, 2012–2017

Subpopulation Identified by the ICF Algorithm	Cohort	No.	% Baseline CVD ^b	% Baseline CHF	No. of Events	Risk %	Crude Risk		IPTW-Weighted Risk		IPTW- and IPCW-Weighted Risk	
							Difference %		Difference %		Difference %	
							RD	95% CI	RD	95% CI	RD	95% CI
Overall population	SGLT2i	10,343	51.1	14.7	425	4.11	-2.53	-3.15, -1.91	-0.77	-1.39, -0.15	-0.78	-1.39, -0.16
Edema	GLP1RA	9,993	53.2	19.3	663	6.63						
	SGLT2i	1,470	69.4	33.3	115	7.8	-5.71	-7.78, -3.64	-4.01	-6.10, -1.92	-3.92	-5.98, -1.87
Male, no edema, no DPP4i	GLP1RA	1,862	70.5	37.2	252	13.5						
	SGLT2i	2,323	51.7	11.8	78	3.4	-2.64	-3.85, -1.42	-1.16	-2.38, 0.05	-1.22	-2.43, -0.02
Male, no edema, DPP4i	GLP1RA	2,286	55.2	16.1	137	6.0						
	SGLT2i	2,195	54.4	12.7	66	3.0	-1.98	-3.37, -0.59	-0.67	-1.99, 0.65	-0.68	2.00, 0.63
Female, no edema, age ≤70 years	GLP1RA	1,283	54.8	16.4	64	5.0						
	SGLT2i	1,760	33.8	8.6	39	2.2	-2.16	-3.26, -1.06	-0.40	-1.52, 0.72	-0.39	-1.51, 0.73
Female, no edema, age >70 years	GLP1RA	2,193	35.9	11.1	96	4.4						
	SGLT2i	2,595	49.1	12.7	127	4.9	0.08	-1.11, 1.28	1.49	0.28, 2.70	1.51	0.31, 2.70
	GLP1RA	2,369	53.0	17.7	114	4.8						

Abbreviations: CHF, congestive heart failure or cardiomyopathy; CI, confidence interval; CVD, cardiovascular disease; DPP4i, dipeptidyl peptidase-4 inhibitor; HHF, hospitalized heart failure; ICF, iterative causal forest; IPCW, inverse-probability of censoring weight; IPTW, inverse-probability of treatment weight; RD, risk difference.

^a Our cohort included 10,343 SGLT2i initiators and 9,993 GLP1RA initiators. Patients were followed from their second prescription until the end of Medicare enrollment, death, or December 31, 2017, whichever came first. A total of 1,854 (9.1%) were censored (as they lost part A or B of Medicare coverage during follow-up). Death was treated as a competing risk by setting the risk for HHF after death to 0 for the 879 (4.3%) patients who died (46). In total, 380 patients (7.82%) were censored and 708 (14.56%) had HHF events. All patients were required to enter the cohort no later than December 31, 2015, so that we could potentially follow patients for a maximum of 2 years (the study ended on December 31, 2017).

^b CVD was defined as major atherosclerotic cardiovascular diseases including myocardial infarction, ischemic heart disease, stroke, cerebrovascular diseases, cardiovascular revascularization procedures, peripheral arterial diseases, and revascularization.

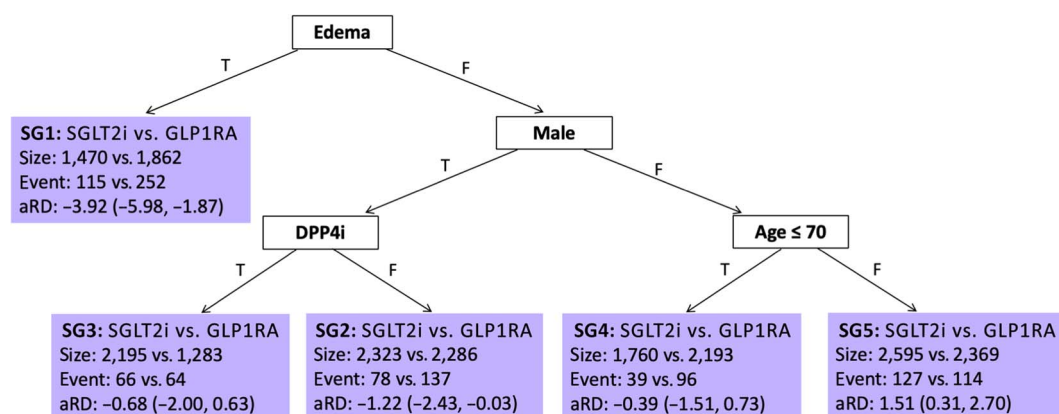


Figure 3. Decision tree from an iterative causal forest (iCF) (with 1,000 iterations) on sodium-glucose cotransporter-2 inhibitors (SGLT2i) vs. glucagon-like peptide-1 receptor agonists (GLP1RA) Medicare new-user cohort and per-subgroup average treatment effect (i.e., conditional average treatment effect). Risk differences adjusted by inverse-probability of treatment and inverse-probability of censoring weighting (on a percentage scale). aRD, adjusted risk difference; DPP4i, dipeptidyl peptidase-4 inhibitor; SG, subgroup.

subgroup identification. To estimate CATEs by RD in each identified subgroup, we applied IPTW (as described in the simulation studies) multiplied by the inverse-probability of censoring weight (IPCW) to control for informative censoring (44). We compared the distribution of 2 important treatment modifiers (45, 47)—established cardiovascular disease (CVD) and congestive heart failure (CHF)—across treatment groups. We also implemented the other ML methods used in the simulation study in PS-matched cohorts (details in Web Appendix 5). For the binary outcome, we used logistic regression with LASSO penalty and aVirtualTwins instead of IT (which only supports continuous outcomes).

All computation for simulations and interaction/subgroup identification was conducted in R (version 4.1; R Foundation for Statistical Computing, Vienna, Austria). In Medicare data, data management and estimation of CATEs was carried out in SAS (version 9.4; SAS Institute, Inc., Cary, North Carolina).

RESULTS

Simulation studies

The mean *P* values of the homogeneity test for scenarios without and with HTEs across 500 simulations for $n = 20,000$ were 0.566 and below 0.001, respectively (Web Table 2). For interaction identification, the iCF and oneCF both outperformed other methods, except for IT, in most scenarios (Figure 2). IT had both the highest specificity and highest sensitivity, followed by the iCF and oneCF. Other methods had poor performance due to false-negative (ROWSi, FindIt) or false-positive (LASSO, FindIt) findings (Web Tables 3–6).

For both subgroup identification and identifying the subgroup with the largest CATE, the iCF outperformed IT for two 2-way interactions or interactions involving ordinal X_2 . While IT outperformed the iCF for 2-way interactions or

interactions involving continuous X_2 . The iCF may have split on the correct variable but often found the wrong splitting value, thus failing to identify the subgroup with the largest CATE. IT performed poorly as it often identified fewer subgroups (Web Table 5). For smaller sample sizes, the iCF performed worse, but its performance relative to IT actually improved. Notably, all tree-based methods had high specificity (≥ 0.8). The average computing time of the iCF (approximately 3 hours) was approximately 5- and 30-fold greater than that of oneCF and IT, respectively (Web Table 7).

Empirical example

The raw CF produced a *P* value of 0.0003 for the homogeneity test and selected 30 important covariates (Web Figure 3, Web Tables 8–9) to develop the iCF. We present estimated CATEs (Table 2) by the subgroups identified from the iCF (Figure 3), which split patients by baseline edema, sex, dipeptidyl peptidase-4 inhibitor use, and age. In each subgroup, the weighted standardized absolute mean differences between SGLT2i and GLP1RA for covariate distribution ranged from 0 to 0.1, with a mean value of 0.03 (Web Table 10). Edema was identified as the most important variable, and the subpopulation with edema had the largest CATE (adjusted RD = -3.92%, 95% confidence interval: -5.98%, -1.87%) and the highest baseline prevalence of CVD (69.4% vs. 70.5%) and CHF (33.3% vs. 37.2%) compared with other subpopulations (Figure 4). The oneCF algorithm identified subgroups by similar features as those used by the iCF except that it missed edema and age. The aVirtualTwins method identified some subgroups with large, but less precise, CATEs defined by CVD and loop diuretics (Figure 4). The logistic regression with LASSO penalty identified interactions of SGLT2i with CHF and home oxygen. ROWSi also identified the SGLT2i:edema interaction (Web Table 11).

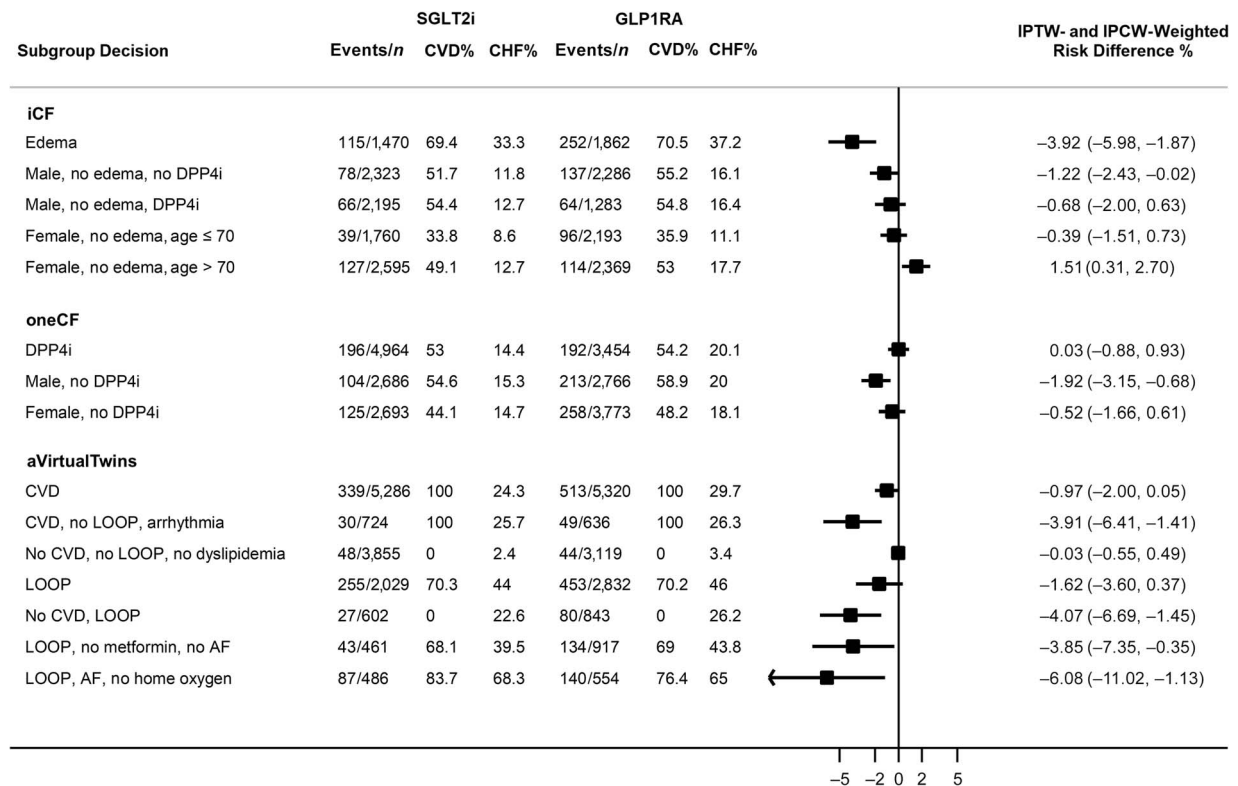


Figure 4. Subgroup decisions and conditional average treatment effects from different forest-based methods, using data from Medicare claims, United States, 2012–2017. Cardiovascular disease (CVD) is defined as major atherosclerotic cardiovascular diseases including myocardial infarction, ischemic heart disease, stroke, cerebrovascular diseases, cardiovascular revascularization procedures, peripheral arterial diseases, and revascularization. The iteration number for the iterative causal forest (iCF) is 1,000. Due to sampling variance, the subgroup decision from one causal forest (oneCF) is not stable. Repeat oneCF on the full cohort many obtain a different subgroup decision. Subgroup decisions from the aVirtualTwins R (R Foundation for Statistical Computing) package (19) are not mutually exclusive. AF, atrial fibrillation; CHF, congestive heart failure or cardiomyopathy; DPP4i, dipeptidyl peptidase-4 inhibitor; GLP1RA, glucagon-like peptide-1 receptor agonists; HHF, hospitalized heart failure; IPCW, inverse-probability of censoring weight; IPTW, inverse-probability of treatment weight; LOOP, loop diuretics; SGLT2i, sodium-glucose cotransporter-2 inhibitors.

DISCUSSION

We developed a new CF-based subgrouping method, the iCF, for large cohorts and non-genomic data with key covariates measured. Unlike other tree-based methods using a pruned decision tree to stratify the population, the iCF first uses a nonparametric approach to develop a few candidate subgroup decisions from different depths, and then fits a parametric model to predict transformed outcomes, formulating subgroup selection into a model selection question.

We recommend the iCF for identifying subgroups in large cohort studies, especially in pharmacoepidemiologic studies using claims data and where most variables are dichotomous, as simulations showed that the iCF performs better in the majority of scenarios for large sample sizes than other methods and successfully prevents false positives. For a continuous variable such as age, the iCF may not accurately identify the splitting value; thus, we recommend categorizing continuous variables before running the iCF. As no method outperforms all others in every scenario, we recommend sensitivity analyses using other methods.

For the empirical study, the subgroups with the largest CATE identified by the iCF or aVirtualTwins could be explained by SGLT2i's beneficial effect on HHF (35–40, 45). Both edema (a fundamental sign of heart failures) and loop diuretics (a key treatment) are proxies for heart failure and thus any of SGLT2i's beneficial effects on HHF can be assumed to be stronger on the RD scale in patients at higher risk for HHF (48, 49). The subpopulation with edema identified by the iCF had the highest CVD and CHF proportions, which is consistent with the study by Htoo et al. (45) suggesting a large CATE in the subgroup with a history of both CVD and heart failure (adjusted RD = -4.77%, 95% confidence interval -8.31%, -1.65%). We note that the iCF did not identify the largest CATE subgroup with a history of heart failure but no CVD, as suggested by Htoo et al. (adjusted RD = -7.38%, 95% confidence interval -12.08, -1.71), which is less precise due to a small number of events ($n = 23$) in SGLT2i initiators (45). Although the algorithm we used to define prevalent heart failure has high specificity (99%), its sensitivity is 80% (45). Heart failure in Medicare claims is not consistently coded, which may explain why

the iCF identified a proxy (edema) rather than heart failure itself.

The iCF provides an efficient subgrouping solution for real-world data without prior knowledge of interactions and seems to outperform oneCF, as shown in our empirical example. We recommend implementing the iCF with a large number of iterations and tree numbers for real-world data with many covariates. It is worth noting that a lack of strong evidence for treatment heterogeneity from the homogeneity test does not mean heterogeneity is absent (24). In such situations, adopting a larger α level (e.g., 0.2) or bypassing the test may reduce false negatives.

The iCF has limitations. First, based on the “no unmeasured confounding” assumption, the iCF should be implemented with epidemiologic study designs that reduce the potential for unmeasured confounding in real-world data (e.g., active-comparator, new-user designs in pharmacoepidemiology). Second, like other tree-based methods, splitting continuous variables may lead to subgroup decisions that are not helpful for clinical decision-making (e.g., age ≤ 70 vs. > 70) (50). Moreover, the iCF may not work well for small data, and estimated CATEs in small subgroups may be unreliable and uninterpretable (21). Third, as the iCF assesses HTEs by the RD scale, further work is needed to subgroup survival data based on the CF (51). Fourth, the iCF may be computationally expensive for big data. Last, real-world data structures are complicated; we tested only a few scenarios in our simulations and implemented the iCF in one empirical example using claims data where only one variable was continuous (age). More empirical studies are needed to test the iCF’s performance in other settings in which we have prior knowledge of HTEs.

CONCLUSION

We developed and assessed the comparative performance of a new CF-based subgrouping method, the iCF, using patient characteristics. Our simulations show that the iCF outperforms other methods in identifying subgroups and the subgroup with the largest CATE in the majority of scenarios using data with varying sample sizes. The iCF is a promising method for identifying subgroups with heterogeneous treatment effects in large cohorts, where the potential for unmeasured confounding can be limited by study design.

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